Exercise 11: UART Transmitter

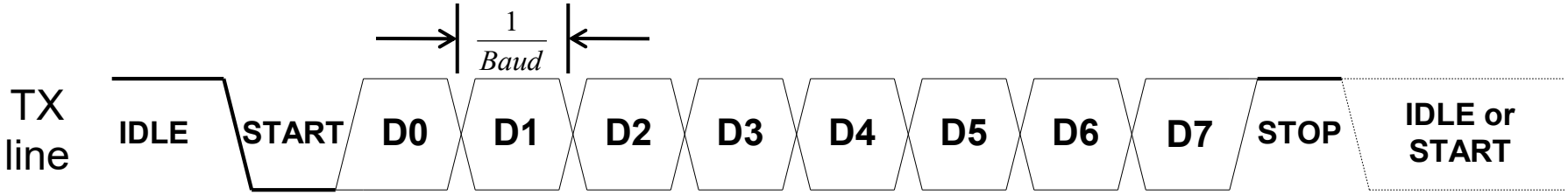
Universal Asynchronous Receiver / Transmitter (UART)

- For **asynchronous serial** communication
- The transmitter (TX) and receiver (RX) have an agreed upon **Baud rate** and **frame format**
  - Standard Baud rates: 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600
  - Frame formats:

1 start bit	5-9 data bits	0 or 1 parity bit	1 or 2 stop bit

- Transmitter sits idle (high) until told to transmit.

- We will use what's commonly referred to as the **8N1** format
  - 8 data bits, no parity bit, 1 stop bit

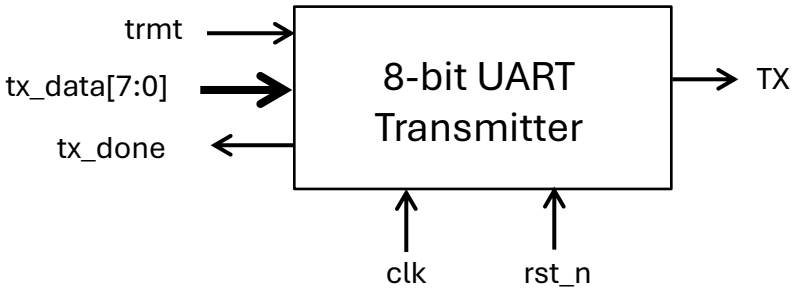


- We will use Baud rate = 9600

Baud rate 9600 with 50MHz clock ➔ 5208 divider ➔ 13-bit

Exercise 11: UART Transmitter

- You are designing the datapath and FSM for this UART transmitter
  - The UART transmitter will receive an 8-bit **tx_data**, and a 1-bit **trmt** signal to tell it to transmit
  - The **tx_data** should be put into the **8N1** frame format, and transmitted on to the **TX** line using a **9600** Baud rate
  - **tx_done** should be asserted once the data has been transmitted



Signal	Dir	Description
clk,rst_n	in	50MHz system clock & active low reset
TX	out	Serial data output
trmt	in	Asserted for 1 clock to initiate transmission
tx_data[7:0]	in	Byte to transmit
tx_done	out	Asserted when byte is done transmitting. Stays high till the next byte transmits.

UART TX Design

Suggested process:

- Think about **what** kind of things need to be done
  → what datapath elements can do that for you?
- Then think about **when** these things need to be done
  → how should the FSM control signals going to the data path?

For example:

- Bits are transmitted 1 bit at a time → Need registers to hold data to be transmitted
- When Baud period is up, transmit next bit → Need counter for Baud rate
- Once all bits are transmitted, assert tx_done → Need counter to keep track of how many bits have transmitted

- Are there different "phases" that the circuit need to work in?
  - Yes. Need to wait for input, and we don't do anything in this phase. Then need to transmit. So we will need to have an FSM.
  - Figure out the datapath elements first before looking at when to change the control signals.

UART TX Design

- Need:
  - Registers to hold data to be transmitted

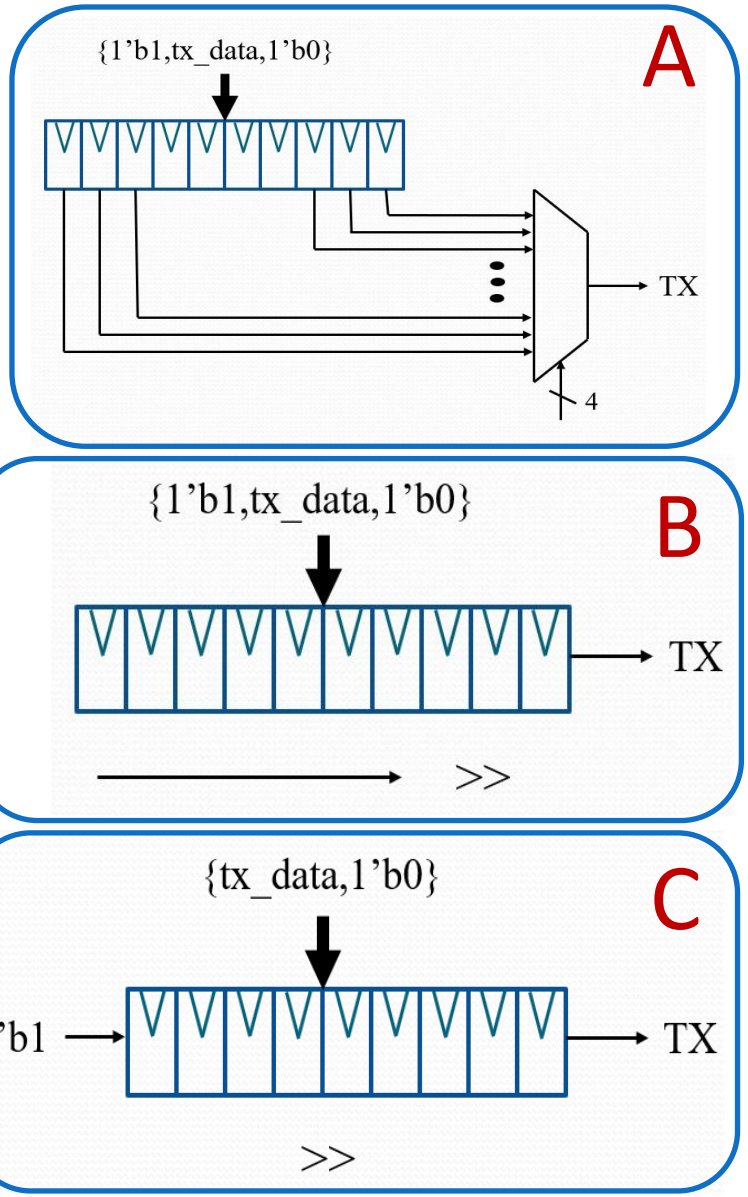
There are a few possible designs:

A. Can have a register load in the frame (all 10 bits) and use a MUX to select which bit to output onto TX

B. Can have a shift register to load in the frame (all 10 bits) and shift right every time. TX is just the value at bit 0.

C. Shift register like B, but can save 1 register for the stop bit and just tie a static 1'b1 (since line is high during idle)

- Should pick the most area efficient / performant design!



UART TX Design

- Need:
  - Registers to hold data to be transmitted
  - Counter for baud rate – sets up when to put data on the line
  - Counter for which bit is being transmitted

- You know how to implement a counter, but how many bits do you need? Do you need a reset? Do you need an enable? Do you need a parallel load? Do you want to count up or count down?

Baud counter

Baud rate = 9600
System clk = 50MHz
→ 50MHz / 9600 = 5208
→ 13-bit counter

- When enabled, keep counting from 0 to 5208
- Send out a signal every time the Baud period is reached

Bit counter

10 bits in a frame to transmit
→ 4-bit counter

- Start from 0, increment every time we transmit a new bit
- Keep track of which bit so we know when the frame is done

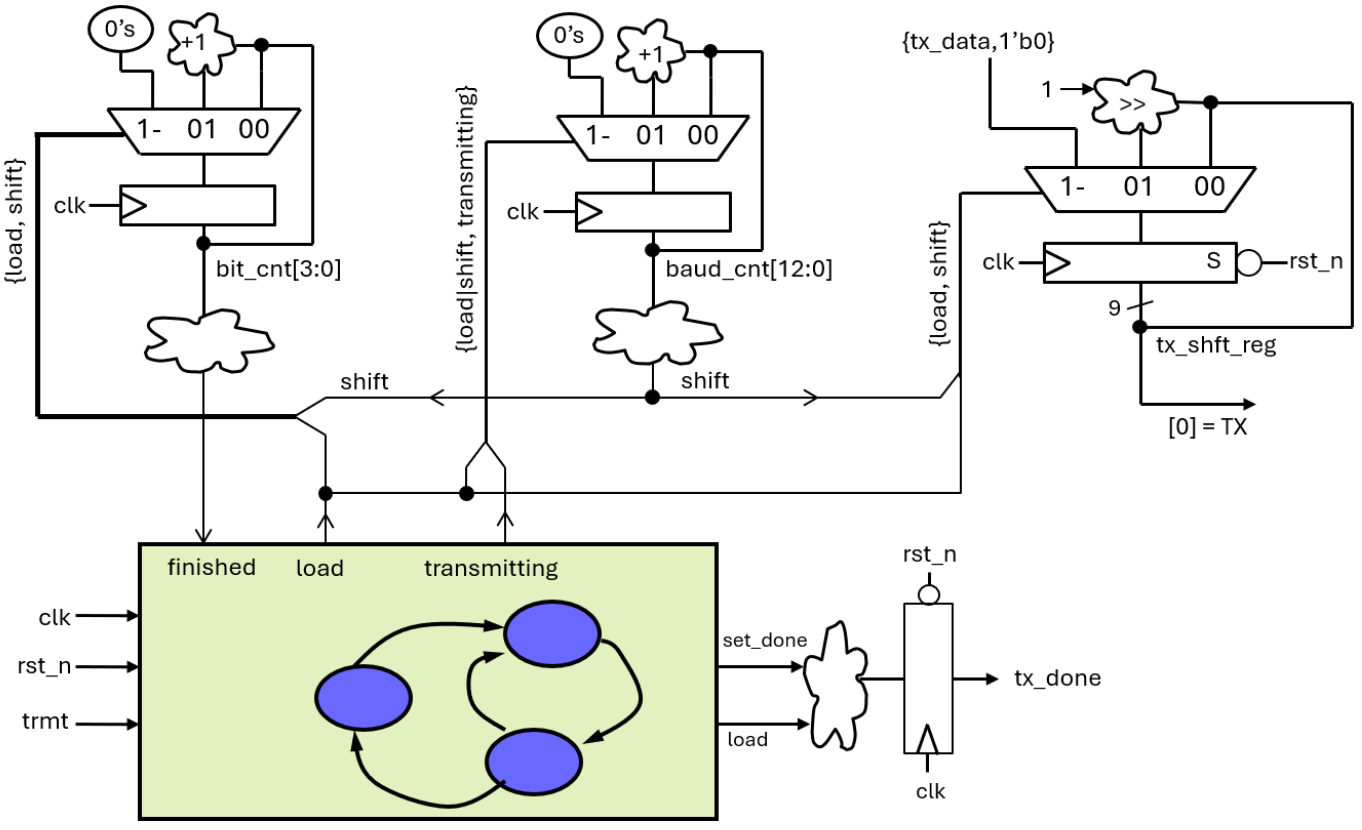
UART TX Design

- Need:
  - Registers to hold data to be transmitted
  - Counter for baud rate – sets up when to put data on the line
  - Counter for which bit is being transmitted
  - FSM to manage control signals – when to do what

Possible topology:

This is recommended. But you are free to use your own design as long as it has an FSM and operates the same.

Remember that the lower the area the better!



Exercise 11: UART Transmitter

- Your design should be called UART_tx
  - Implement the datapath
  - Implement the FSM
    - What is the minimum number of states you need?
- You should follow the FSM coding style introduced in the lecture videos

- Once your design is done, make a simple testbench for it.
  - Just instantiate your transmitter and send a few bytes. Verify correct functionality (including baud rate) by looking at the waveform.
  - We will test it again later when you have your UART receiver as well.

- **Submit: UART_tx.sv**